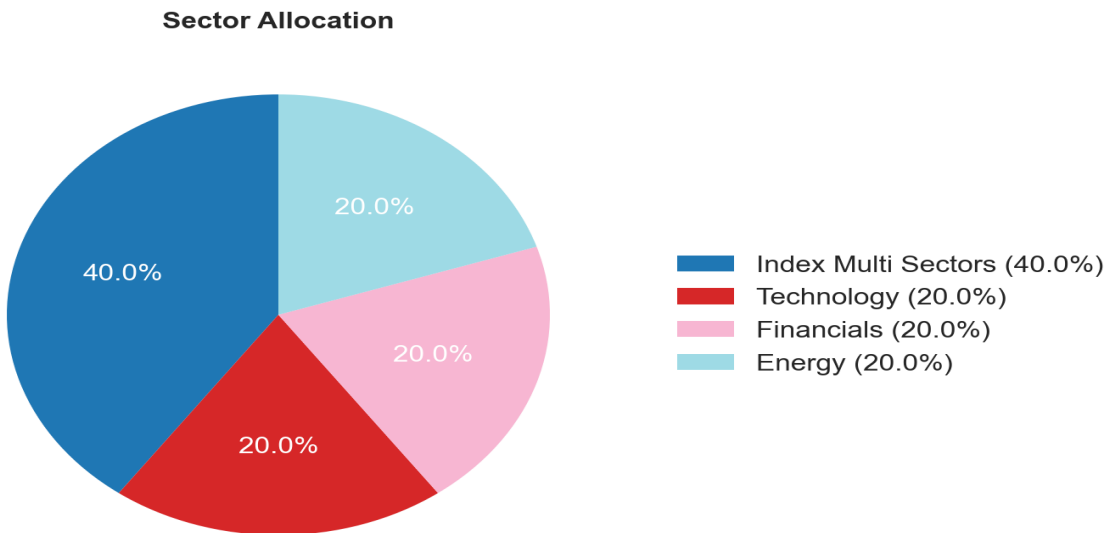


Portfolio rotation report

Date of extraction	2025-12-07 22:08
Data window	2019-12-30 → 2024-10-21
Frequency	Weekly — 252 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+384.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	QQQ, SPY, XLE, XLF, XLK
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 20 weeks volatility, 24 weeks return, 48 weeks momentum

1. Portfolio Composition

- XLK (Technology Select Sector SPDR Fund — Technology)
 - XLF (Financial Select Sector SPDR Fund — Financials)
- XLE (Energy Select Sector SPDR Fund — Energy)
 - SPY (SPDR S&P; 500 ETF — Index Multi Sectors)
 - QQQ (Invesco QQQ Trust — Index Multi Sectors)



Macro Profile Insight

Macro coverage

- Partial coverage: several macro themes remain unrepresented.
- Broad balance across market capitalizations: diversification achieved across the size spectrum.

Structural properties

- Concentration risk: portfolio heavily exposed to etf instruments.
- Liquidity stance: pockets of thin/illiquid names; robust median dollar volume.
- Market-neutral beta exposure: close alignment with index dynamics.
- Volatility profile skewed to dynamic/speculative names.

Additional comments

- Significant sector gaps: broad portions of the economy remain uncovered.

2. Sector Performance Visualization



Top Positive Correlations
100.0% of portfolio

Pair	Correlation
QQQ / XLK	0.97
QQQ / SPY	0.91
SPY / XLK	0.91
SPY / XLF	0.84
XLE / XLF	0.7

Overall, average cross-sector correlation was 0.68, ranging from 0.33 to 0.97. This suggests very high correlation — sectors behave almost as a single macro block. Correlation dispersion remains meaningful, indicating both tightly and weakly linked sectors. Strongest co-movement between QQQ and XLK (0.97); weakest between QQQ and XLE (0.33). Market-driven regime: macro factors dominate individual sector dynamics.

3. Key Performance Indicators

Ticker	R (%)	V (%)	RAR	M (%)	Signal	Regime	Liquidity	Commentary
QQQ	0.85	3.19	0.266	0.01	neutral	subdued	liquid	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
SPY	0.78	2.26	0.346	0.34	mildly coherent	subdued	liquid	mixed short-term dynamics, signal remains coherent, but with limited conviction, low volatility — calm market tone, with ample liquidity.
XLE	0.24	4.06	0.06	0.06	neutral	subdued	liquid	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
XLF	0.99	2.04	0.484	0.87	mildly coherent	subdued	thin	positive performance with solid momentum, signal remains coherent, but with limited conviction, low volatility — calm market tone, under limited liquidity.
XLK	1.05	4.23	0.249	-0.02	neutral	normal	thin	positive returns but fading momentum, signal unclear or in transition, normal volatility regime, under limited liquidity.

Top by Average Return

Ticker	AvgReturn(%)
XLK	1.05
XLF	0.99
QQQ	0.85
SPY	0.78
XLE	0.24

Top by Volatility

Ticker	Volatility(%)
XLK	4.23
XLE	4.06
QQQ	3.19
SPY	2.26
XLF	2.04

Top by Momentum

Ticker	Momentum(%)
XLF	0.87
SPY	0.34
XLE	0.06
QQQ	0.01
XLK	-0.02

3.b Rotation & Stress Metrics

Ticker	ΔReturn (pp)	Momentum Ratio	Signal	Volatility Ratio	Beta	ADV10 USD	Interpretation
QQQ	0.39	0.02	0.01	1.10	1.18	53.05 M	stable performance, momentum fading, volatility normalizing, amplifying market trends, under thin liquidity regime.
SPY	0.30	0.55	0.22	1.20	1.00	72.91 M	stable performance, momentum fading, volatility normalizing, amplifying market trends, under thin liquidity regime.
XLE	0.37	0.29	0.047	1.44	0.80	28.61 M	stable performance, momentum fading, volatility expanding, amplifying market trends, under thin liquidity regime.
XLF	0.48	1.18	0.399	1.07	0.84	36.84 M	stable performance, momentum stable, volatility normalizing, amplifying market trends, under thin liquidity regime.
XLK	0.60	-0.04	-0.019	1.08	1.33	16.33 M	stable performance, momentum fading, volatility normalizing, amplifying market trends, under thin liquidity regime.

■ Light-impact observations

For **QQQ**, **SPY**, **XLE**, **XLF**: *Volatility compression* — subdued volatility; a breakout setup may be forming. (avg ann. return 0.7%, momentum 0.3%, vol 2.9%)

For **XLK**: *Range / noise* — no clear direction; rotating momentum and mid-range volatility. (avg ann. return 1.1%, momentum -0.0%, vol 4.2%)